



EXAMINATION NO.: _____
THE MALAWI NATIONAL EXAMINATIONS BOARD
2019 MALAWI SCHOOL CERTIFICATE OF EDUCATION EXAMINATION

CHEMISTRY

Subject Number: M038/I

Thursday, 27 June

Time Allowed: (2 hours)
8:00 – 10:00 am

PAPER I

THEORY

(100 marks)

Instructions

1. This paper contains 12 printed pages. Please check.
2. Fill in your Examination Number at the top of each page.
3. This paper contains two sections, A and B. In Section A there are ten short answer questions while in Section B there are three restricted essay questions.
4. Answer all the thirteen questions in the spaces provided.
5. Use of electronic calculators is allowed.
6. The maximum number of marks for each answer is indicated against each question.
7. In the table provided on this page, tick against the number of the question you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
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9			
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11			
12			
13			

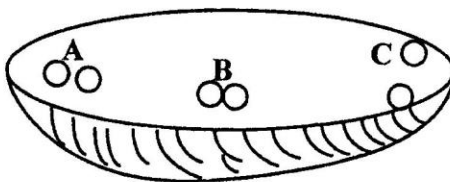
Section A (70 marks)

Answer **all** the questions in this section.

1. a. State the difference between “accuracy” and “precision”.

(2 marks)

- b. **Figure 1** is a diagram of a container with molecular models in various positions. The models are supposed to be at the centre of the container.

**Figure 1**

Using letters A, B and C, identify the molecular models that have-

- (i) good precision but poor accuracy

(1 mark)

- (ii) both poor precision and accuracy

(1 mark)

- (iii) both good precision and accuracy.

(1 mark)

- c. In an experiment, 136g of gaseous ammonia (NH_3) reacted with excess oxygen (O_2) to produce nitric acid (HNO_3) and water (H_2O).

- (i) Write a balanced equation for the reaction.

(3 marks)

Continued/...

1c. (continued)

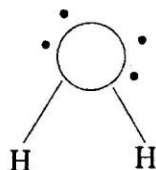
- (ii) How much nitric acid (HNO_3) could be produced from this reaction?
(RAM: N=14, H = 1, O = 16)

(3 marks)

2. a. Work out the number of nitrogen atoms in 8 molecules of urea
 $\text{CO}(\text{NH}_2)_2$.

(2 marks)

- b. Complete the equation and label the dative bond.



+

 H^+ 

(2 marks)

- c. Explain each of the following statements.

- (i) Ionic compounds are soluble in water.

(3 marks)

- (ii) Graphite conducts electricity.

(3 marks)

Continued/...

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2. (continued)

- d. Explain how a polar bond is formed.

(4 marks)

3. a. Give any **one** use of electron configuration.

(1 mark)

- b. (i) Why is nitrogen gas inert at room temperature?

(2 marks)

- (ii) Explain any **one** way in which inertness of nitrogen is important.

(3 marks)

Continued/...

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3. (Continued)

- c. The Table below shows electron configuration and relative abundance of isotopes of an element.

Isotope	Electron configuration	Relative Abundance
X - 28	2 - 8 - 4	$\frac{8}{10}$
X - 29	2 - 8 - 4	$\frac{1}{10}$
X - 30	2 - 8 - 4	$\frac{1}{10}$

- (i) To which group of the periodic table does element X belong?

(1 mark)

- (ii) Element X reacts with element Y whose valency is 2.
Write the chemical formula of the product.

(2 marks)

4. a. A hydrocarbon contains 85.7% carbon and 14.3% hydrogen by mass.
Write the molecular formula of the hydrocarbon if its relative formula mass is 56. (RAM: C=12, H=1)

(7marks)

Continued/...

4. (continued)

- b. Why is burning of plastics **not** recommended as a method of disposal?

(2 marks)

5. Figure 2 is a graph showing results of experiments A and B on one of the factors affecting rate of reaction.

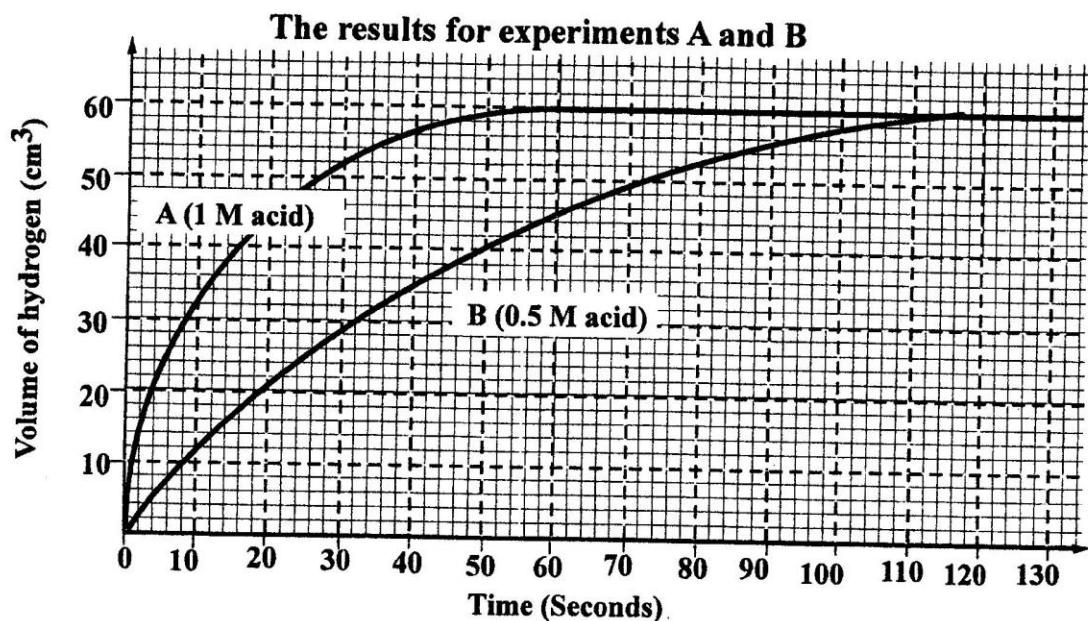


Figure 2

Adapted from Complete Chemistry by Rose pp 122

- a. State the factor affecting the rate of reaction which was being investigated.

(1 mark)

- b. Which reaction is faster than the other?

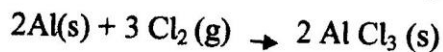
(1 mark)

- c. Give a reason for the answer in 5.b.

(1 mark)

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6. Aluminium reacts with chlorine gas according to the following equation:



- a. Write oxidation and reduction half reactions

Oxidation: _____

(2 marks)

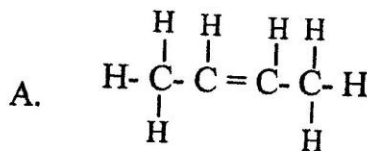
Reduction: _____

- b. Calculate the volume of 8M KNO_3 stock solution to be diluted to 400ml so that the final concentration is 0.2M.

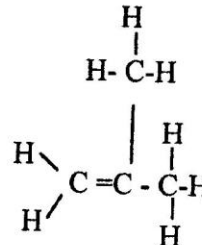
(2 marks)

7. Figure 3 shows different structures of an organic compound.

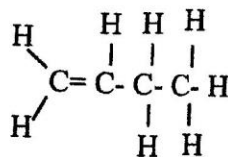
(3 marks)



B.



C.



D.

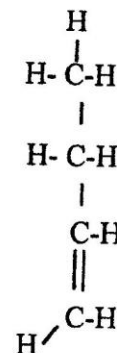


Figure 3

- a. Identify conformers.

(1 mark)

- b. Which of these structures has the lowest boiling point?

(1 mark)

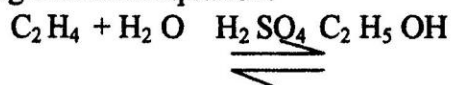
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7. (Continued)

- c. Give a reason for the answer in 7(b).

(2 marks)

8. a. Ethene (
- C_2H_4
-) and water (
- H_2O
-) react to form ethanol (
- C_2H_5OH
-) according to the following chemical equation.



- (i) Name the type of reaction in the equation.

(1 mark)

- (ii) Give the function of
- H_2SO_4
- in the reaction.

(1 mark)

- b. (i) Complete the following equation:



(2 marks)

- (ii) What is the function of
- $Cr_2O_7^{2-}$
- in the reaction?

(1 mark)

- (iii) Name the products in the equation.

(2 marks)

Continued/...

9. a. Why is argon gas obtained before oxygen gas during separation of air into its component gases?

(1 mark)

- b. Describe how carbon dioxide is removed from air.

(2 marks)

10. a. Mention any **one** type of wastes based on degradability.

(1 mark)

- b. Explain how burning of fuel causes global warming.

(2 marks)

Continued/...

Answer all the questions in this section.

-
-
-
-

b. With the aid of chemical equations, explain how ozone layer is depleted.

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Continued/...

(10 marks)
Continued/...

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13. With the aid of a well labelled diagram, describe how copper can be refined using **electrolysis**. In your description, include half equations at the cathode and anode.

(10 marks)

END OF QUESTION PAPER

NB: This paper contains 12 printed pages.